ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

TARGET : JEE (Main + Advanced) : 2021

CLASS : ENTHUSIAST (XII)

TEST TYPE: ONLINE | **DATE : 29.11.2020** | **PATTERN : JEE ADVANCED**

Time : 3 Hours | **Maximum Marks : 198**

SRG TEST - 8

INSTRUCTIONS

READ THE INSTRUCTIONS CAREFULLY

QUESTION PAPER FORMAT AND MARKING SCHEME

1. The question paper has **three parts** : **PHYSICS**, **CHEMISTRY** and **MATHEMATICS**. Each part has **THREE SECTIONS**.
2. Carefully read the instructions given at the beginning of each section.
3. **Section I(i):** This section contains **FOUR** questions. The answer to each question is a **ONLY ONE CORRECT OPTION.**

   Marking scheme : +3 for correct answer, 0 if not attempted and –1 in all other cases.
4. **Section I(ii):** This section contains **SIX** questions. The Answer to each question is a **ONE OR MORE THAN ONE CORRECT OPTIONS**.

Marking scheme :	+4	for correct answer, 0 if not attempted and –2 in all other cases.
Partial Marks :	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks :	+2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks :	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
5. **Section I(iii):** This section contains **THREE** questions based on paragraph. The answer to each question is a **ONLY ONE CORRECT OPTION.**

   Marking scheme : +3 for correct answer, 0 if not attempted and –1 in all other cases.
6. **Section II:** This section contains **SEVEN** questions. The answer to each question is a **NUMERICAL VALUE**.

   Marking scheme : +3 correct answer, 0 if not attempted and 0 in all other cases.

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SOME USEFUL CONSTANTS

Atomic No. H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17,Br = 35, Xe = 54, Ce = 58,

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- **Boltzmann constant** $k = 1.38 \times 10^{-23}\ JK^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11}\ N\text{–}m^2\ kg^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8\ ms^{-1}$
- **Stefan–Boltzmann constant** $\sigma = 5.67 \times 10^{-8}\ Wm^{-2}\text{–}K^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3}\ m\text{–}K$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7}\ NA^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34}\ J\text{–}s$

ENTHUSIAST COURSE

SRG TEST- 7 **DATE : 22 - 11 - 2020**

SYLLABUS

PHYSICS : FULL SYLLABUS

CHEMISTRY : FULL SYLLABUS

MATHS : FULL SYLLABUS

PART-1 : PHYSICS

SECTION–I(i) : (Maximum Marks : 12)

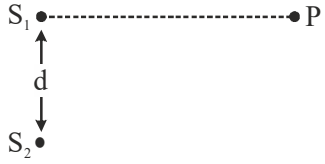
- This section contains **FOUR** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : –1 In all other cases7

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1. Consider two coherent monochromatic (wavelength λ) sources S_1 and S_2 separated by distance d. The ratio of intensity of S_1 and that of S_2 (which is responsible for interference at point P, where detector is located) is 4. The distance of point P from S_1 is (if the resultant intensity at point P is equal to $\frac{9}{4}$ times of intensity of S_1)

[Given $\angle S_2S_1P$ is 90°] (d > 0, n is a positive integer)

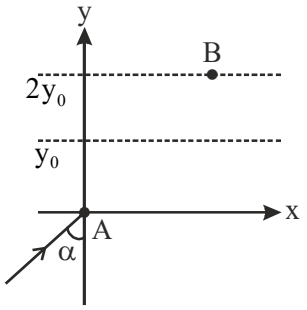


(A) $\frac{d^2 - n^2\lambda^2}{2n\lambda}$ (B) $\frac{d^2 + n^2\lambda^2}{2n\lambda}$

(C) $\frac{n\lambda d}{\sqrt{d^2 - n^2\lambda^2}}$ (D) $\frac{2n\lambda d}{\sqrt{d^2 - n^2\lambda^2}}$

2. The refractive index of space changes with y, whose function is given as

$$\mu(y) = \begin{cases} 1 & \text{if} \quad y \le 0 \\ ky^2 & \text{if} \quad 0 \le y \le 2y_0 \\ \frac{\ell}{y^2} & \text{if} \quad 2y_0 \le y \le 4y_0 \end{cases}$$



where k and ℓ are positive constant. When a light ray is incident at point A as shown in figure. When the ray reaches to point B, it becomes parallel to xz-plane. Choose the correct statement.

(A) α is independent of k (B) α is independent of ℓ

(C) α is independent of both k and ℓ (D) None of these

3. The space has electromagnetic field which varies with time whose variation is given as

$$\vec{B} = \begin{cases} -B_0\hat{k} & \text{if} \quad 0 \le t \le \frac{\pi m}{qB_0} \\ -B_0\hat{j} & \text{if} \quad \frac{\pi m}{qB_0} \le t \le \frac{2\pi m}{qB_0} \\ 0 & \text{if} \quad \frac{2\pi m}{qB_0} \le t \le \infty \end{cases} \text{ and } \vec{E} = \begin{cases} 0 & \text{if} \quad 0 \le t \le \frac{2\pi m}{qB_0} \\ -E_0\hat{k} & \text{if} \quad \frac{2\pi m}{qB_0} \le t \le \infty \end{cases}$$

A charge particle having mass m and positive charge q is given velocity $v_0\hat{i}$ at origin at t = 0 sec. The

coordinate of point on xy plane when it is again passing through xy-plane for the first time is :

(A) $\left(\frac{2mv_0}{q}\sqrt{\frac{v_0}{E_0B_0}}, \frac{2mv_0}{qB_0}, 0\right)$ (B) $\left(\frac{2mv_0}{q}\sqrt{\frac{v_0}{E_0B_0}}, \frac{mv_0}{qB_0}, 0\right)$

(C) $\left(\frac{mv_0}{q}\sqrt{\frac{v_0}{E_0B_0}}, \frac{2mv_0}{qB_0}, 0\right)$ (D) $\left(\frac{mv_0}{q}\sqrt{\frac{v_0}{E_0B_0}}, \frac{mv_0}{qB_0}, 0\right)$

4. A thin string with linear density μ is joined to a thick string with linear density 2μ. A incident pulse is sent down the thin string toward the thick string and eventually creates reflected and transmitted pulses. Which of following is true ?

(A) The reflected and transmitted pulses are both inverted
(B) Neither the reflected nor transmitted pulses are inverted
(C) The reflected pulse is inverted, but the transmitted pulse is not inverted
(D) The transmitted pulse is inverted, but the reflected pulse is not inverted

SECTION–I(ii) : (Maximum Marks : 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

5. In the shown figure (1) and (2), capacitors are in the steady state. Charging batteries are removed and switches S_1 and S_2 are closed at time t = 0. The plot of $\ell n I$ (I is the current in the resistor) in against time t in the resistors R_1 and R_2 are shown by the graphs 1 and 2 respectively in the figure (3). Choose the correct option(s).

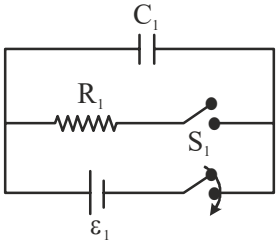


Figure (1)

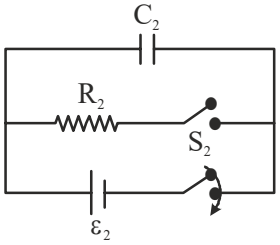


Figure (2)

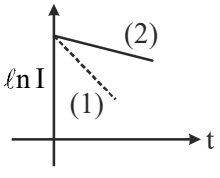


Figure (3)

(A) If $\varepsilon_1 = 2\varepsilon_2$, R_1 must be equal to $2R_2$ (B) If $R_1 = R_2$, C_1 must be equal to C_2

(C) If $C_1 = C_2$, R_1 must be less than R_2 (D) R_1C_1 is equal to R_2C_2

6. X-rays from a tube with a target A of atomic number Z shows strong K line for target A and weak K lines for impurities. The wavelength of K_α lines is λ_z for target A and λ_1 and λ_2 for two impurities. If $\frac{\lambda_z}{\lambda_1} = 4$ and $\frac{\lambda_z}{\lambda_2} = \frac{1}{4}$ and taking screening constant of K_α lines to be unity, select the correct statement(s)

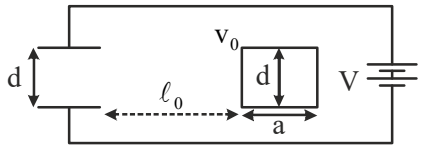
(A) The atomic number of first impurity is $2Z - 1$

(B) The atomic number of first impurity is $2Z + 1$

(C) The atomic number of second impurity is $\frac{(Z+1)}{2}$

(D) The atomic number of second impurity is $\frac{Z}{2}+1$

7. A parallel plates capacitor is made of square conducting plates of side a and the separation between plates is d. The capacitor is connected with battery of emf V volt as shown in the figure. There is a dielectric slab of dimension $a \times a \times d$ with dielectric constant k. At $t = 0$, dielectric slab is given velocity v_0 towards capacitor as shown in the figure. (Neglect the effect of gravity and electrostatic force acting on the dielectric when dielectric is out side of capacitor. Also ignore any type frictional force acting on the dielectric during its motion). Let the x be the length dielectric inside the capacitor at t = t sec. $[\ell_0 >> a]$



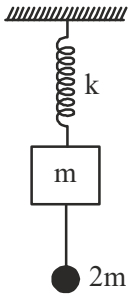
(A) Motion of dielectric slab is periodic but not simple harmonic motion

(B) Motion of dielectric slab is simple harmonic motion

(C) At any time, the slope of graph of total energy verses x is twice the slope of graph of potential energy verses x.

(D) The value of maximum energy stored in the system is $\frac{1}{2}mv_0^2 + \frac{\varepsilon_0 a^2 V^2}{2d}(2k-1)$

8. A bob of mass 2m hangs by a string attached to the block of mass m of a spring block system. The whole arrangement is in a state of equilibrium. The bob of mass 2m is pulled down slowly by a distance x_0 and released.



(A) For $x_0 = \frac{3mg}{k}$, maximum tension in string is 4mg

(B) For $x_0 = \frac{3mg}{k}$, minimum tension in string is mg

(C) Frequency of oscillation of system is $\frac{1}{2\pi}\sqrt{\frac{k}{3m}}$, for all non-zero values of x_0

(D) The motion will remain simple harmonic for $x_0 \le \frac{3mg}{k}$

9. The equation of a process of diatomic gas is $P^2 = \alpha^2 V$ where α is a positive constant. Then choose the INCORRECT option(s).

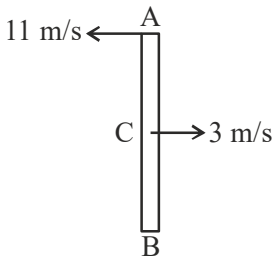
(A) Work done by gas for a temperature change T is $\frac{2}{3}\alpha nRT$

(B) The change in internal energy is $\frac{5}{2}nRT$ for a temperature change T

(C) The specific heat for the process is $\frac{19}{9}R$

(D) The change in internal energy for a temperature change T is $\frac{5}{2}\alpha nRT$

10. A uniform rod AB of length 7m is undergoing combined rotational and translational motion on smooth horizontal surface such that, at some instant of time, velocities of its end point A and centre C are both perpendicular to the rod and opposite in direction, having magnitude 11 m/s and 3 m/s respectively as shown in the figure. Now consider that no external force and torque is acting on the rod. Then select CORRECT options.



(A) Magnitude of acceleration of point A is 56 m/s^2
(B) Magnitude of acceleration of point B is 56 m/s^2
(C) At the instant shown in the figure acceleration of point C is zero
(D) Angular velocity of the rod is 4 rad/s

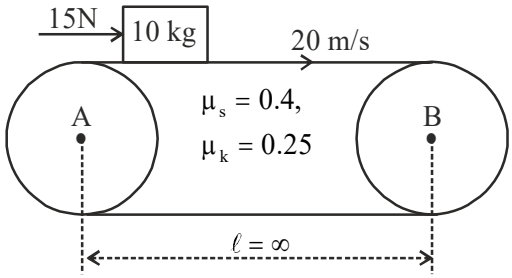
SECTION–I(iii) : (Maximum Marks : 9)

- This section contains **ONE** paragraphs.
- Based on paragraph, there will be **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened.
Zero Marks : –1 In all other cases.

Paragraph for Questions 11 to 13

A block of mass 10 kg is put gently on a belt-conveyor system of infinite length at t = 0 sec, which is moving with constant speed 20 m/sec rightward at all time, irrespectively of any situation by means of a motor-system as shown in the figure. A constant force of magnitude 15 N is applied on the block continuously during its motion.



11. The nature of friction force acting on the block of mass 10 kg during its motion is

(A) Kinetic only (B) Static only

(C) Some time kinetic some time static (D) Can't be predicted

12. Work done by the kinetic friction on the block of mass 10 kg is

(A) 1250 Joule (B) 2500 Joule

(C) –1250 Joule (D) None of these

13. The magnitude of acceleration of the block of mass 10 kg at t = 6 sec is

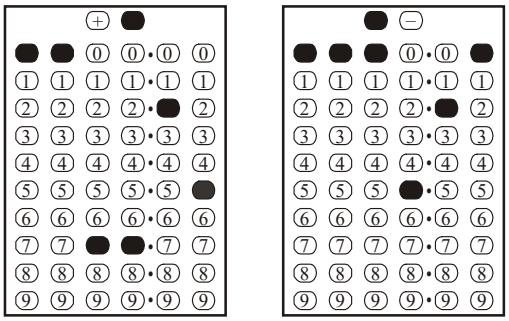
(A) 4 m/s^2 (B) 3 m/s^2

(C) 2 m/s^2 (D) None of these

SECTION-II : (Maximum Marks: 21)

- This section contains **SEVEN** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:

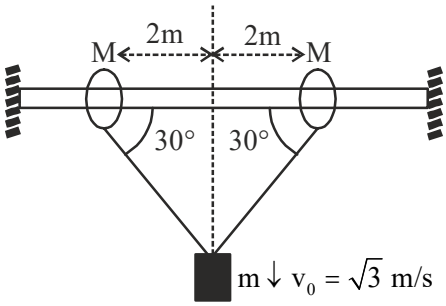
Full Marks : +3 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

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14. The pitch of a screw gauge is 1 mm and there are 50 divisions on its cap. When nothing is put in between the studs, 44th division of the circular scale coincides with the reference line and zero of main scale is not visible. When a plate is placed between the studs, the main scale reads 5 divisions and the circular scale reads 25 divisions. The true thickness of plate is $a \times 10^{-5}$ m, find the value of a.

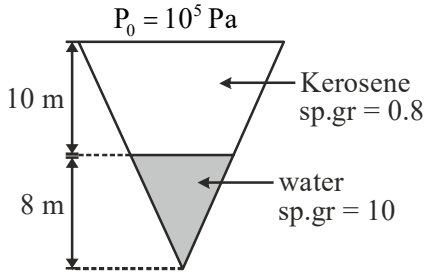
15. Two rings, each of mass M = 100 gm are constrained to move along a fixed horizontal rod. An ideal string is connected with rings and a small m = 200 gm is connected to the mid point of string. At any moment m is moving downward with velocity $\sqrt{3}$ m / s. The speed of ring of M at the same moment is



16. Pressure variation due a sonic wave propagating along positive x-direction is given by the equation $\Delta P(x) = \Delta P_0 \sin[2\pi(\upsilon t - x)]$. One end of an open organ pipe is closed by a cap and held at rest with its axis parallel to x-axis and the open end at the origin. What should be the minimum value of frictional force between the cap and the pipe so that pipe will remain closed. Given $\Delta P_0 = 2\times10^3\ N/m^2$ and cross sectional area of the tube is equal to $5\times10^{-4}\ m^2$,

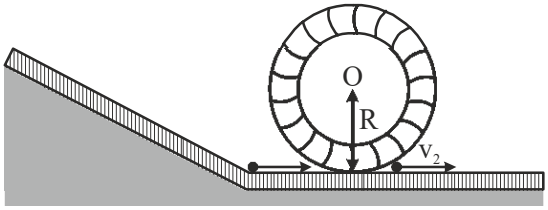
17. In a slow reaction, heat is being evolved at a rate about 10mW in a liquid. If the heat were being generated by the decay of ^{32}P, a radioactive isotope of phosphorus that has half-life of 14 days and emits only beta-particles with a mean energy of 700 KeV, estimate the number of ^{32}P atoms in the liquid. Express your answer in form $A\times10^{15}$ of and fill A in answer sheet. [Take : ln 2 = 0.675]

18. Shown in figure, a conical container of half-apex angle 37° filled with certain quantities of kerosene and water. The force exerted by the water on the kerosene is $A\times10^5$ N fill value of A in answer sheet, upto two decimal places. (Take atmospheric pressure, 10^5 Pa)



19. The half-life of substance X is 45 years, and it decomposes to substance Y. A sample from a meteorite was taken which contained 2% of X and 14% of Y by quantity of substance. If substance Y is not normally found on a meteorite, what is the approximate age of the meteorite?

20. Water flowing along an open channel drives an undershot water wheel of radius 2 m (figure). The water approaches the wheel with a speed of 5.0 m/s and leaves with a speed of 2.5 m/s; the amount of water passing by is 20 kg per second. At what rate does the water deliver angular momentum to the wheel (in N-m) ?



Undershot water wheel

PART-2 : CHEMISTRY

SECTION–I(i) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : –1 In all other cases

1. Among the following which one is more soluble in H_2O

(A) CsF (B) RbF
(C) NaF (D) LiF

2. Wave number of first line in Lyman series of hydrogen spectrum is equal to wave number of which one of the following lines:

(A) First line of Balmer series of He^+ (B) Second line of Paschen series of Li^{2+}
(C) Third line of Paschen series of Li^{2+} (D) First line of Lyman series of Li^{2+}

3. $CO_{2(g)} + C_{(s)} \rightleftharpoons 2CO_{(g)}$ $K_p = 3$bar at 727°C. If initial pressure of CO_2 is 0.48 bar. The partial pressure of CO_2 at equilibrium is

(A) 0.20 bar (B) 0.30 bar
(C) 0.15 bar (D) 0.33 bar

4. Which of the following elements has least electron affinity

(A) Tellurium (B) Sulphur
(C) Oxygen (D) Selenium

SECTION–I(ii) : (Maximum Marks : 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : –2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

5. Among the following salt solutions which give sulphide precipitate when H_2S gas is passed after the addition of NH_4OH

(A) Mn^{2+} (B) Pb^{2+}

(C) Ni^{2+} (D) Cu^{2+}

6. Aspirin and oil of winter green can be distinguished by

(A) Neutral $FeCl_3$ (B) aq. $NaHCO_3$

(C) aq.NaOH (D) Br_2/CCl_4

7. Which of the following reaction/s is/are correct

(A) Monochlorination of 2-Methyl butane gives 2-Chloro-3-methylbutane as major product

(B) Hydrogenation of cis-2-hexene gives a racemic mixture

(C) Addition of excess HCl to actylene gives ethyldenedichloride

(D) Addition of HBr to cis-2-butene gives racemic mixture

8. Extraction of copper from copper pyrites involves

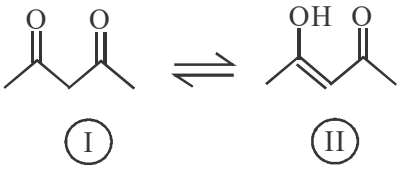
(A) Concentration of the powdered ore by froth flotation

(B) Roasting of the ore in a reverbaratory furnace

(C) Roasting of the copper-matte with silica to remove iron in the form of iron silicate slag

(D) Leaching of the powdered ore with KCN in presence of air

9. Choose the correct statements with respect to following equilibrium



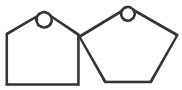
(A) % of form I increases in water

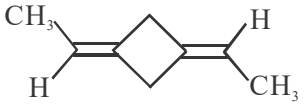
(B) % of form I decreases in water

(C) % of form II increase in ether

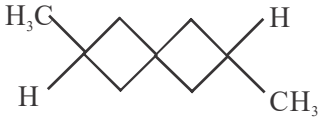
(D) % of form I increases with increase in temperature in ether

10. Which of the following is/are able to exhibit optical isomerism?

(A) 

(B) 

(C) $CH_2 = C = CH_2$

(D) 

SECTION–I(iii) : (Maximum Marks : 9)

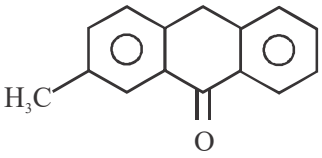
- This section contains **ONE** paragraphs.
- Based on paragraph, there will be **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened.

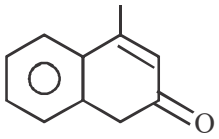
Zero Marks : –1 In all other cases.

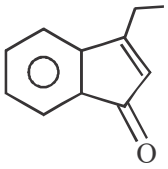
Paragraph for Questions 11 to 13

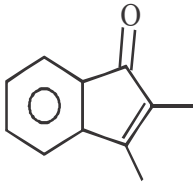
A hydrocarbon P($C_{11}H_8$) on treatment with $HgSO_4$aq. H_2SO_4 gives Q($C_{11}H_{12}O_2$). P on treatment with B_2H_6, THF and then with H_2O_2/ NaOH gives R which is an isomer of Q. Q gives yellow precipitate with I_2/NaOH, where as R do not give haloform reaction. R forms silver mirror on treatment with ammonical silver nitrate solution. P on boiling with alkaline $KMnO_4$ solution forms $C_6H_6O_4$, which on heating with P_2O_5 gives S($C_8H_4O_3$). S on treatment with toulene in presence of anhydrous $AlCl_3$ gives T ($C_{15}H_{12}O_3$) as the major product. T is reduced by Zn-Hg/*HCl* and product obtained gives the following compound on reaction with HF.

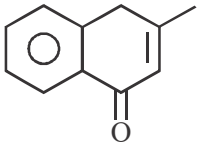


11. Which of the following products are formed when Q is treated with aqueous NaOH?

(a) 

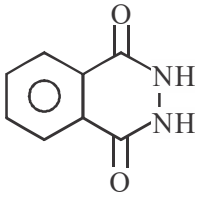
(b) 

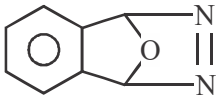
(c) 

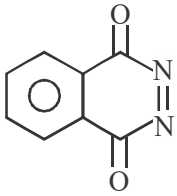
(d) 

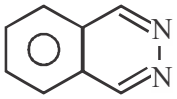
(A) a and d
(B) a, b & c
(C) b and c
(D) a, b, c and d

12. S on heating with hydrazine hydrochloride gives

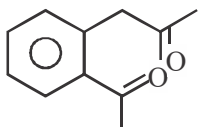
(A) 

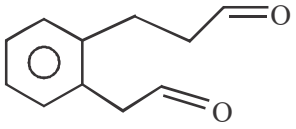
(B) 

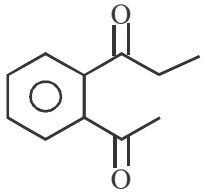
(C) 

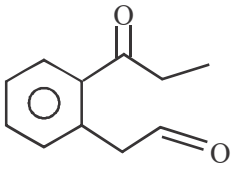
(D) 

13. The structure of R is

(A) 

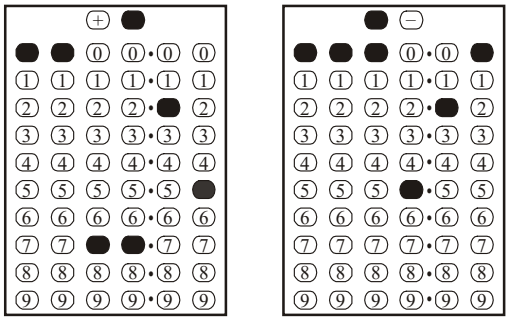
(B) 

(C) 

(D) 

SECTION-II : (Maximum Marks: 21)

- This section contains **SEVEN** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.
  For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +3 If ONLY the correct numerical value is entered as answer.
  Zero Marks : 0 In all other cases.

14. Aqueous solutions of sodium iodate and sodium iodide react in acidic medium to give iodine, the total number of NaI molecules involved in the balanced chemical reaction is _____

15. A solution of 15.168 g of Se_x in 200 gm of benzene showed a depression in freezing point of 0.6°C. If the molal depression constant of benzene is 5 K Kg mol^{-1} and atomic weight of selanium is 79 u, the value of x is _____

16. Find out the value of expression $|x+y|$ for complex ion $\left[Co(EDTA)\right]$. Where, x = Total number of six membered ring of y = Total number of five membered rings having oxygen donor atoms only.

17. Solubility of $Fe(OH)_2$ ($k_{sp} = 10^{-24}$) in a buffer solution containing 0.1 M BOH and 0.1 M B^+ is 10^{-12}. Calculate pKa of $B^{\oplus}$.

18. All allotropic forms of boron contain B_{12} icosohedral units. If there are 'p' faces, 'q' corners and 'r' edges in the icosohedron then $\frac{(p+q-r)}{2}$ is

19. The standard reduction potential of a silver chloride electrode is 0.2 V and that of a silver electrode is 0.79V. If the maximum amount of Ag*Cl* that can dissolve in 10^6 Litres of a 0.1 M $AgNO_3$ solution is y

millimoles, then the value of y is _____

20. 0.2 moles of TlBr($k_{sp} = 10^{-6}$) is added to 1 litre of a saturated solution of Hg_2Br_2 ($k_{sp} = 4\times10^{-24}$). The resultant solution has $2^x \times 10^{-3y}$ M mercurous ion. The value of (x + y) is

PART-3 : MATHEMATICS

SECTION–I(i) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : –1 In all other cases

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1. Let $A=\begin{pmatrix}1 & 3 & 5\\ 4 & 2 & 6\\ 7 & 1 & 3\end{pmatrix}$; $B=\begin{pmatrix}3 & 6 & 7\\ 6 & 3 & 8\\ 7 & 8 & 3\end{pmatrix}$

Let $ABA^T=\begin{pmatrix}a_1 & a_2 & a_3\\ a_4 & a_5 & a_6\\ a_7 & a_8 & a_9\end{pmatrix}$

Then $(a_2-a_4)^2+(a_3-a_7)^2+(a_6-a_8)^2$ will be equal to

(A) 0 (B) 13
(C) 78 (D) 65

2. $\int_1^5 (x-1)(x-2)(x-3)(x-4)(x-5)\,dx$ lies in

(A) [0, 5] (B) [4, 9]
(C) [10, 15] (D) [– 5, 0)

3. $\int \frac{(3x^2+3x+1)\,dx}{x^6+3x^5+3x^4+x^3} = f(x)+k$. Then f(1) = ?

(A) $-\frac{1}{8}$ (B) $-\frac{7}{8}$
(C) $-\frac{3}{8}$ (D) $-\frac{5}{8}$

4. The minimum value of $\sum_{k=1}^{5} k\,|x-k|$ for all $x \in R$ is

(A) 5 (B) 10
(C) 15 (D) 20

SECTION–I(ii) : (Maximum Marks : 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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5. $\int_0^{\pi} (|\sin x| + |\cos x| + ||\sin x| - |\cos x||)dx$ equals to

(A) $\sqrt{2}$ (B) $2\sqrt{2}$

(C) $3\sqrt{2}$ (D) $4\sqrt{2}$

6. Let $f(x) = \left|||x|-3|-2\right|-1\right|$ where $x \in R$.

Let a be the number of points where f(x) is discontinuous

b be the number of points where f(x) is non differentiable

c be the number of local minimas of f(x)

d be the number of local maximas of f(x)

Then a + b + c + d is equal to

(A) 22 (B) 24

(C) 26 (D) 28

7. The equation $ln\, x = kx^4$ has atleast one solution. Then which of the following values of k are possible

(A) $-\frac{1}{e}$ (B) $\frac{1}{(4e)}$

(C) $\frac{1}{(5e)}$ (D) $\frac{1}{(10e)}$

8. The number of solutions of the equation $3x = 100\{x\}$ is

(A) 34 (B) 31

(C) 32 (D) 33

9. Consider a function $f : [-3, 1] \to [-11, 16]$ such that $f(x) = x^3 - 12x$. The function f can be classified as

(A) One-one (B) Onto

(C) Into (D) Many-one

10. Compute $\int_0^{10} l\,n\left(x^7+1\right)dx + \int_0^{l\,n(10000001)} \sqrt[7]{e^x-1}\,dx$.

(A) $l\,n\left(10^7+1\right)$ (B) $10\,l\,n\left(10^7+1\right)$

(C) $10^7\,l\,n\left(10^7+1\right)$ (D) $100\,l\,n\left(10^7+1\right)$

SECTION–I(iii) : (Maximum Marks : 9)

- This section contains **ONE** paragraphs.
- Based on paragraph, there will be **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened.

Zero Marks : –1 In all other cases.

Paragraph for Questions 11 to 13

Sixteen players S_1, S_2, ..., S_{16} play in a tournament. They are divided into 8 pairs at random. From each pair a winner is decided on the basis of a game played between the two players of the pair. Assume that all the players are of equal strength.

11. The probability that exactly one of the two players S_1 & S_2 is among the eight winners is

(A) $\frac{4}{15}$ (B) $\frac{7}{15}$

(C) $\frac{8}{15}$ (D) $\frac{9}{15}$

12. What is the probability that all of S_1, S_2, S_3, S_4 are among the winners ?

(A) $\frac{1}{{}^{16}C_8}$ (B) $\frac{{}^{12}C_4}{{}^{16}C_8}$

(C) $\frac{{}^{8}C_4}{{}^{16}C_8}$ (D) $\frac{4}{{}^{16}C_8}$

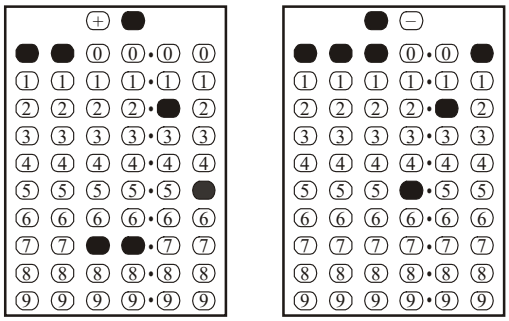
13. If the knockout procedure is repeated until a final winner emerges, what is the probability that S_2 wins the tournament ?

(A) $\frac{1}{16}$ (B) $\frac{1}{8}$

(C) $\frac{3}{16}$ (D) $\frac{1}{4}$

SECTION-II : (Maximum Marks: 21)

- This section contains **SEVEN** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.
  For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +3 If ONLY the correct numerical value is entered as answer.
  Zero Marks : 0 In all other cases.

---

14. If sum and the difference of the roots of the characteristic equation of a square matrix of order 2×2 are 4 and 2 respectively then the determinant of the matrix will be

15. Define functions f_0, f_1, ... , f_n, ... to satisfy the following relations : $f_0 = 2$, $\frac{d}{dx}f_n = f_{n-1}$. If for $n \geq 1$, $f_n(0) = 0$, compute $\sum_{n=0}^{\infty} f_n(l\,n(2019))$.

16. Let A be the area of the region bounded by the curve $5x^2 + 6xy + 2y^2 + 7x + 6y + 6 = 0$, then [A] is equal to

17. If $\begin{bmatrix} 4a^2 & 4a & 1 \\ 4b^2 & 4b & 1 \\ 4c^2 & 4c & 1 \end{bmatrix}\begin{bmatrix} f(-1) \\ f(1) \\ f(2) \end{bmatrix} = \begin{bmatrix} 3a^2+3a \\ 3b^2+3b \\ 3c^2+3c \end{bmatrix}$, $f(x)$ is a quadratic function and its maximum value occurs at a point V. A is a point of intersection of y = f(x) with x-axis and point B is such that chord AB subtends a right angle at V. Let A be the area enclosed by f(x) and chord AB. Find [A] (where $[\cdot]$ denotes the greatest integer function)

18. A tangent is drawn to the curve $y = x^3 + x^2 + x + 1$ at A where the curve intersects the line y = 4. This tangent intersect the given curve at $B(x_1, y_1)$. Then $|x_1 + y_1|$ is

19. Let $f : R \to R$ be a continuous and differentiable function. f(x) = 0 has 10 real roots. Then the minimum number of real roots of the equation $f' \cdot f'' + f\,f''' = 0$ is

20. Consider the differential equation $y'' - 5y' + 6y = 0$. It can be converted to a linear differential equation of the form $y' - 3y = c \cdot f(x)$. Then f(*l*n 3) is

ENTHUSIAST COURSE

SRG TEST- 8	DATE : 29 - 11 - 2020
SYLLABUS	
PHYSICS : FULL SYLLABUS	
CHEMISTRY : FULL SYLLABUS	
MATHS : FULL SYLLABUS	